

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)
Homework 1 – GEOS-Chem & Air Quality Simulation over China
Total Score: 75

Learning Objectives:

- Understand the fundamental workflow of GEOS-Chem Classic, a global 3D chemical transport model used to simulate atmospheric composition. Drive a nested GEOS-Chem simulation from a global run's boundary conditions and extract output from both.
- Locate and interpret the primary NetCDF output files and ensure the simulation completed successfully with physically plausible results.
- Locate a geographic region as GEOS-Chem grid-relative indices, correctly read the variables, and convert to observation units.
- Compute bias, correlation, and diurnal metrics for common air pollutants, and explain each discrepancy in terms of emissions, chemistry, meteorology, or observation density.

Part 0. Basics of GEOS-Chem and Nested Grid Simulation

Please read the following website to study the output files and the related settings: <https://geos-chem.readthedocs.io/en/stable/gcclassic-user-guide/diag-outputs-hist.html>. Then, answer the following questions.

Question 1 (2 pts). By reading `SpeciesConc.log`, you will see the line containing `'time = UNLIMITED ; // (1 currently)'`, which describes how many records are in the file. Which setting controls **how often a record is written**?

file:

setting:

For the `GEOSChem.SpeciesConc.*.nc4` files, they are output day by day. Which setting controls **how long a single file spans before a new file is created**?

file:

setting:

Question 2 (1 pt). If we want to see `'time = UNLIMITED ; // (24 currently)'`, and still output day by day. How to revise the settings?

file:

previous settings:

revised settings:

Question 3 (2 pts). Read the `SpeciesConc.log`, you can see a lot of variables named as `'SpeciesConcVV_*'`, such as `SpeciesConcVV_02`, `SpeciesConcVV_N2`... Which setting in which file leads to it?

file:

setting:

`SpeciesConc.fields`:

If we only need 1 variable: `SpeciesConcVV_S02`, how to revise the setting?

```
# file:
# previous settings:
# revised settings:
```

Question 4 (2 pts). The output files includes the collections , AdvFluxVert, AerosolMass, Aerosols, DryDep, JValues, Metrics, SpeciesConc, StateMet? Which setting, and in which file, controls whether a collection is output? What does each collection mean?

```
# file:
# setting:
# meanings of each collection:
```

If we only need to output 2 types: *SpeciesConc*.nc4 and the Restart files. How do we revise the settings?

```
# file:
# previous settings:
# revised settings:
```

Question 5 (1 pt). Where is the output file 'GEOSChem.Restart.20190703_0000z.nc4' located at?
Directory:

Question 6 (1 pt). If we want to run the simulation from 20190703 at 12:00 PM Beijing time to 20190705 at 8:00 AM Central European time, how should we revise the settings?

```
# file:
# previous settings:
# revised settings:
```

Question 7 (1 pt). The global simulation runs at $4^\circ \times 5^\circ$ resolution and the nested simulation runs at $0.5^\circ \times 0.625^\circ$ resolution.

- (a) Roughly how many times finer is the nested grid than the global grid, per grid cell?
- (b) At $\sim 40^\circ\text{N}$, approximate the physical size of one grid cell in each case. Use $1^\circ \approx 111$ km for latitude and $1^\circ \approx 85$ km for longitude at 40°N .

Question 8 (3 pts). The nested run needs boundary conditions from the global run. The file that tells GEOS-Chem where to find them is HEMCO_Config.rc.

- (a) Which switch must be set to true to enable reading boundary conditions? Give the exact name and value.
- (b) Which block in HEMCO_Config.rc specifies the file path for the boundary condition files? Give the exact file pattern used.
- (c) Why is the nested domain set up to be so much larger than the region of interest? Give one scientific reason.

Question 9 (2 pts). Both `Restart` and `BoundaryConditions` are collections in `HISTORY.rc`, and both are `.nc4` files that the model reads in. But they play completely different roles.

- (a) In one sentence each, state the purpose of a restart file and a boundary condition file.
- (b) Which of the two is read only once, at the start of a simulation? Which is read continuously throughout the run?

Question 10 (2 pts). Why can a nested $0.5^\circ \times 0.625^\circ$ run resolve city-scale pollution that a $4^\circ \times 5^\circ$ global run cannot? Give one concrete example.

Question 11 (2 pts). In one sentence, what do the boundary conditions represent, and why can the nested run not produce them itself?

Part 1. Guided Setup for a Nested Run over Beijing

We consider the Beijing region, using the bounding box $\text{LON} = [116, 117]$, $\text{LAT} = [39, 40]$.

I. Setup for Global Simulation

[HISTORY.rc](#)

For this file, we configure the output collections to save space. Keep only the `'Restart'` collection, the `'SpeciesConc'` collection, the `'AerosolMass'` collection, and the `'BoundaryConditions'` collection uncommented.

```
#####
# %%%% COLLECTION NAME DECLARATIONS %%%%
#
# To enable a collection, remove the "#" character in front of its name. The
# Restart collection should always be turned on.
#
# NOTE: These are the "default" collections for GEOS-Chem, but you can create
# your own customized diagnostic collections as well.
#####
COLLECTIONS: 'Restart',
             #'Metrics',
             'SpeciesConc',
             #'AdvFluxVert',
             'AerosolMass',
             #'Aerosols',
             #'Budget',
             #'CloudConvFlux',
             #'ConcAboveSfc',
             #'ConcAfterChem',
             #'DryDep',
             #'JValues',
             ...
             #'WetLossConv',
             #'WetLossLS',
             'BoundaryConditions',
..
```

```

=====
# %%% THE SpeciesConc COLLECTION %%%
#
# GEOS-Chem species concentrations (default = all species)
#
# Available for all simulations
#
# Concentrations may be saved out as SpeciesConcVV [v/v dry air] or
# SpeciesConcMND [molec/cm3]
#=====
SpeciesConc.template:      '%y4%m2%d2_%h2%n2z.nc4',
SpeciesConc.frequency:    00000000 010000
SpeciesConc.duration:     00000001 000000
SpeciesConc.mode:         'time-averaged'
SpeciesConc.fields:       'SpeciesConcVV_S02      ',
                          'SpeciesConcVV_N02      ',
                          'SpeciesConcVV_O3       ',
                          'SpeciesConcVV_CO       ',
                          '#SpeciesConcVV_?ALL?    ',
                          '#SpeciesConcMND_?ALL?    ',
::

```

```

=====
# %%% THE AerosolMass COLLECTION %%%
#
# Aerosol and PM2.5 mass
#
# Available for full-chemistry and aerosol-only simulations
#=====
AerosolMass.template:      '%y4%m2%d2_%h2%n2z.nc4',
AerosolMass.frequency:    00000001 000000
AerosolMass.duration:     00000001 000000
AerosolMass.mode:         'time-averaged'
AerosolMass.fields:       'PM25                  ',
                          '#AerMassASOA          ',
                          '#AerMassBC           ',
                          '#AerMassHMS          ',
                          '#AerMassINDIOL        ',
                          '#AerMassLVOCOA         ',
                          '#AerMassNH4           ',
                          '#AerMassNIT           ',
                          '#AerMassOPOA          ',
                          '#AerMassPOA           ',
                          '#AerMassSAL           ',
                          '#AerMassS04           ',
                          '#AerMassSOAGX          ',
                          '#AerMassSOAIE          ',
                          '#AerMassTSOA          ',
                          '#BetaNO              ',
                          '#PM10                ',
                          '#TotalBiogenicOA        ',
                          '#TotalOA              ',
                          '#TotalOC              ',
::

```

```

=====
# %%% THE BoundaryConditions COLLECTION %%%
#
# GEOS-Chem boundary conditions for use in nested grid simulations
#
# Available for all simulations
#
# NOTE: Do not change the BoundaryConditions.frequency setting below,
# because GEOS-Chem nested-grid simulations expect to read boundary
# condition data at 3 hour intervals.
#=====
BoundaryConditions.template: '%y4%m2%d2_%h2%n2z.nc4',
BoundaryConditions.frequency: 00000000 030000
BoundaryConditions.duration:  00000001 000000
BoundaryConditions.mode:      'instantaneous'
BoundaryConditions.LON_RANGE: 112.0 120.0,
BoundaryConditions.LAT_RANGE: 35.0 42.0,
BoundaryConditions.fields:    'SpeciesBC_?ADV?      ',
::

```

[geoschem_config.yml](#)

```
#####
# Simulation settings
#####
simulation:
  name: fullchem
  start_date: [20190701, 000000]
  end_date: [20190703, 000000]
  root_data_dir: /lustre/project/eesc4550/GEOS-Chem/ExtData
  met_field: MERRA2
  species_database_file: ./species_database.yml
  species_metadata_output_file: OutputDir/geoschem_species_metadata.yml
  verbose:
    activate: false
    on_cores: root      # Allowed values: root all
  use_gcclassic_timers: true
```

For this file, we just have to change the `start_date` and `end_date`.

Now, we are ready to run a global simulation.

II. Setup for Nested Simulation

[Creating a Nested-Grid Simulation Run Directory](#)

```
$ /lustre/project/eesc4550/<your-computing-id>/GEOS-Chem/my_code_dir/run  # or whatever you named the code directory
$ ./createRunDir.sh
```

We use the finer grid resolution of $0.5^{\circ} \times 0.625^{\circ}$. **Remember to compile and copy code files.**

[HISTORY.rc](#)

Same as the global simulation, except that we no longer save the 'BoundaryCondition' collection.

[geoschem_config.yml](#)

Also configure `nested_simulation`, set `activate` to `true` and the values of the respected `latitude` and `longitude` of the region. Make sure that the box is larger than the region itself.

```
#####
# Simulation settings
#####
simulation:
  name: fullchem
  start_date: [20190701, 000000]
  end_date: [20190703, 000000]
  root_data_dir: /lustre/project/eesc4550/GEOS-Chem/ExtData/
  met_field: MERRA2
  species_database_file: ./species_database.yml
  species_metadata_output_file: OutputDir/geoschem_species_metadata.yml
  verbose:
    activate: false
    on_cores: root      # Allowed values: root all
  use_gcclassic_timers: false

#####
# Grid settings
#####
grid:
  resolution: 0.5x0.625
  number_of_levels: 47
  longitude:
    range: [ 110.0, 133.0]
    center_at_180: true
  latitude:
    range: [ 30.0, 46.0]
    half_size_polar_boxes: true
  nested_grid_simulation:
    activate: true
    buffer_zone_NSEW: [3, 3, 3, 3]
```

HEMCO Config.rc

Turn the GC_BCs option to true and update the BC_ entry to point to the boundary condition files generated in the global simulation.

```
# ExtNr ExtName          on/off Species
0      Base             : on      *
# ----- RESTART FIELDS -----
--> GC_RESTART           :      true
--> GC_BCs               :      true # <== make sure this is true
--> HEMCO_RESTART        :      true
...
# --- GEOS-Chem boundary condition file ---
#=====
(((GC_BCs
* BC_ /lustre/project/eesc4550/<your-computing-id>/GEOS-
Chem/GC_rundirs/gc_4x5_47L_merra2_fullchem_benchmark/OutputDir/GEOSChem.BoundaryConditions.$YYYY$MM$DD_$HH$MNz.nc4
SpeciesBC_?ADV? 1900-2100/1-12/1-31/* CYS xyz 1 * - 1 1
)))GC_BCs
```

Finally, use a restart file generated by high-resolution grid settings by creating a symbolic link to the file `/lustre/project/eesc4550/shared/nested/GEOSChem.Restart.20190701_0000z.nc4` in the `Restarts` directory. Now, we are ready to run a nested simulation for the region we care.

Part 2. Conceptual and Analytical Questions

Question 12 (2 pts). Identify the nearest grid cell indices in the **nested** grid, for the Beijing region.

Question 13 (4 pts). Extract `SpeciesConcVV_03`, `SpeciesConcVV_N02` concentration over the Beijing box, first time step, on the surface. Extract the surface `PM25` concentration from the `AerosolMass` file over the same box. Report the spatial mean in $\mu\text{g}/\text{m}^3$.

Part 2. Observation Data Processing

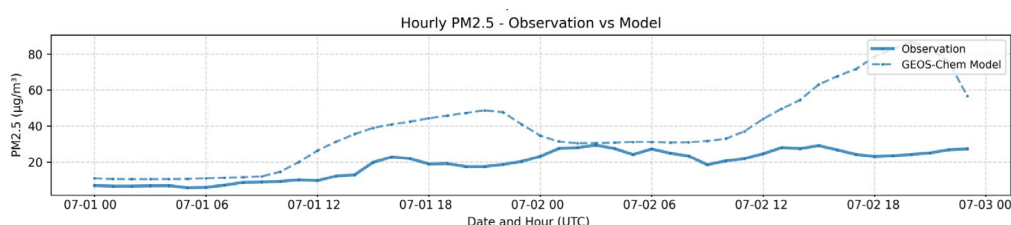
Question 14 (5 pts). The observations are in $\mu\text{g}/\text{m}^3$ for gases. Write the conversion formula from $\mu\text{g}/\text{m}^3$ to ppbv at 25 °C and 1 atm. Give the multiplicative factor for SO_2 , NO_2 , O_3 , and CO .

Question 15 (5 pts). Inspect the file `/lustre/project/eesc4550/shared/china_sites_data/china_sites_20190701.csv`. How many rows does it have? What are the unique values in the type column? Which pollutant has the most missing data? Which has the least?

Part 3. Output Extraction and Data Analysis

Question 16 (10 pts). Use the observation data under the directory `/lustre/project/eesc4550/shared/china_sites_data`. Produce a 5-panel figure, with one pollutant per panel, depicting the observations (solid line) vs. model (dashed line) over the 48-hour window. Specify the stations you used for the observation data.

Example ($\text{PM}_{2.5}$):



Question 17 (10 pts). Compute the statistics for each pollutant, in the observation's native units.

Pollutant	obs_range	model_range	obs_mean	model_mean	bias = mean(model – obs)	<i>r</i>
PM _{2.5}						
SO ₂						
NO ₂						
O ₃						
CO						

Question 18 (10 pts). Select **TWO** pollutants from SO₂, NO₂, O₃, and CO. For **each pollutant**, write one sentence describing the **observations and model simulations**: the range and the temporal evolution. Follow the style of the example below.

Example (PM_{2.5}): “Observations range 5.7 – 29.3 µg/m³, with a gradual increase from low values on July 1 (5 – 10 µg/m³) to a peak of ~ 29 µg/m³ on July 2 morning, then remaining elevated (20 – 29 µg/m³). Model ranges 10.5 – 86.7 µg/m³, showing a sharp rise on July 1 afternoon (reaching ~ 48 µg/m³ by evening) and an even higher peak on July 2 evening (~ 87 µg/m³).”

Question 19 (10 pts). For **each pollutant you described in Question 18**, write one sentence describing the **differences**: the bias direction, magnitude, and any diurnal phase mismatch, and **possible reasons** for the discrepancy. Each reason must name a specific mechanism, e.g. emissions, chemistry, meteorology, or observation density.

Example (PM_{2.5}): “The model consistently overestimates PM2.5, particularly on July 2 where observed peaks are ~ 29 µg/m³ but the model reaches ~ 87 µg/m³; the diurnal pattern also differs, as observations show a broad morning peak on July 2, while the model peaks in the evening. Possible reasons include overestimated primary aerosol emissions or secondary formation, underestimated deposition, or boundary layer dynamics that trap pollutants too efficiently in the model.”

Submission Guideline: To complete the homework, please submit TWO files:

- **hw1_<your_id>.pdf**, which contains the answers to all the **19** questions;
- **hw1_<your_id>.zip**, which contains the Python scripts to produce the included plots and scripts that perform data analysis.

If any AI tool was used in writing or code generation, please include a short note on AI use.

----- End of Homework 1 -----